

FACULYZE

A Hybrid Feature Selection Framework for Faculty Evaluation Analytics

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INTRODUCTION

- Great teachers shape the future, but measuring faculty performance fairly is tough.
- Traditional feedback often misses patterns and insights.
- FacuLyze applies Data Warehousing & Data Mining to feedback, uncovering hidden trends and classifying faculty into:
 - Needs Improvement
 - Good
 - Moderate
- This ensures fair evaluation, growth opportunities, and better student outcomes.

PROBLEM STATEMENT

Universities collect large amounts of student evaluation feedback every semester, but most institutions fail to extract actionable insights from this data. Traditional analysis is manual, inconsistent, and unable to identify factors that strongly influence student satisfaction and faculty performance.

Goal:

Build a machine learning system that predicts student-rated performance levels based on course and feedback data, using an optimized hybrid feature-selection model.

MOTIVATION

- Continuous improvement in teaching
- Recognize & reward faculty
- Decision support for administrators
- Better student outcomes

LITERATURE SURVEY



No.	Title & Year	What They Did (Algorithms/Methods)	How Faculyze is different
1	Zhao et al. (2020) – A Hybrid Feature Selection Method Based on Correlation Matrix	Proposed a hybrid FS using correlation matrix + search strategies. Improved accuracy on high-dimensional datasets but evaluated only one hybrid combination.	Faculyze systematically evaluates all 11 hybrid ratios (0:10 → 10:0) and enforces strict disjoint features, giving deeper insight into method-wise contribution.
2	Wang et al. (2019) – Hybrid FS using IG + SVM	SVM combined Information Gain (filter) with SVM-based wrapper. Focused on improving classification but did not explore multiple FS ratios or disjoint feature sets.	Faculyze introduces a ratio-based hybrid framework and ensures no overlap between Chi-Square and DT-selected features, enabling fair comparison.

LITERATURE SURVEY



No.	Title & Year	What They Did (Algorithms/Methods)	How Faculyze is different
3	Islam et al. (2020) – Hybrid FS in Medical Data	Developed a hybrid FS for medical diagnosis datasets using statistical + model-based selectors. Explored hybrid accuracy but only in binary medical data.	Faculyze applies hybrid FS in multi-class faculty evaluation, a novel domain, and performs complete ratio testing across 11 combinations.
4	Abu Tair & El-Halees (2012) – Mining Educational Data to Improve Students' Performance	Used classical ML (DT, NB, KNN) for student performance prediction; no hybrid FS, no disjoint subset enforcement, no ratio exploration.	Faculyze is the first to combine hybrid FS + disjoint enforcement in educational analytics and shifts focus from student performance to faculty performance evaluation.

BENEFITS AND ISSUES

BENEFITS

- Recognize outstanding faculty
- Support continuous improvement
- Enhance teaching quality

ISSUES

- Subjectivity in feedback
- Unstructured & scattered data
- Lack of fairness in evaluations
- Limited insights from averages
- Scalability challenges

OBJECTIVES

- Identify the most influential features determining student performance ratings
- Develop a hybrid feature-selection approach using filter (Chi-square) and embedded (Decision Tree) methods
- Ensure disjoint selection (no overlapping features) to avoid redundancy
- Analyze performance across all combinations from 0:10 $\rightarrow$ 10:0 feature ratios
- Evaluate the best feature set using accuracy and confusion matrices
- Build an interpretable, reproducible ML pipeline for institutions

TOOLS AND TECHNIQUES USED

TOOLS

- Python - Programming Language
- Jupyter Notebook - Execution Environment
- Pandas , Numpy - Data Handling
- Matplotlib - Visualization
- Scikit-learn - Evaluation Metrics

TECHNIQUES

- Data preprocessing (cleaning, encoding labels)
- Feature selection (Chi Square + Decision Tree)
- Decision tree
- Model evaluation (accuracy, precision, recall, F1-score)

DATASET DESCRIPTION

Dataset: Turkiye Student Evaluation Dataset

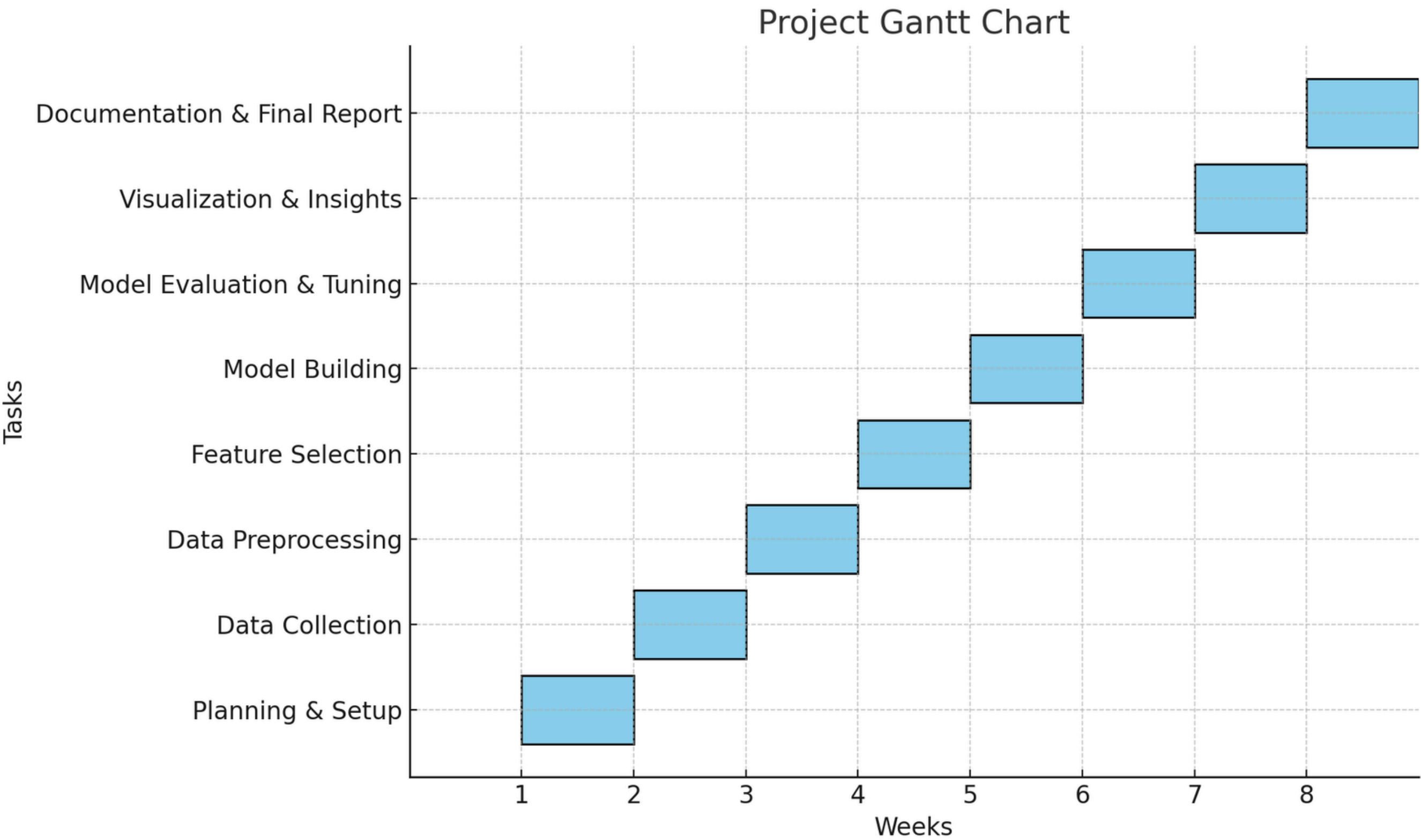
- Contains 5820 student evaluation entries
- Features include:
 - Instructor ID, Class ID
 - Attendance
 - Difficulty
 - Number of repeats
 - 28 student rating questions (Q1–Q28)

Target variable created:

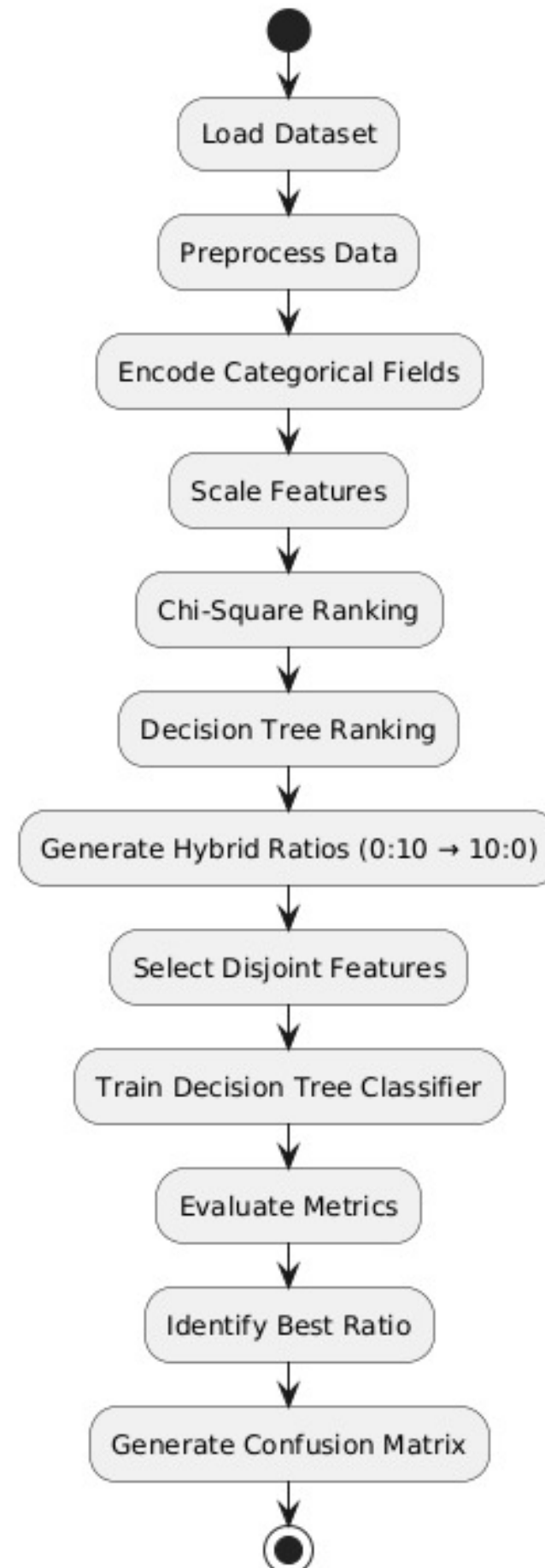
- Performance:
 - Needs Improvement
 - Moderate
 - Good
 (converted from avg_rating)

	A	B	C	D	E
1	instr	class	nb.repeat	attendance	difficulty
2	1	2	1	0	4
3	1	2	1	1	3
4	1	2	1	2	4
5	1	2	1	1	3
6	1	2	1	0	1
7	1	2	1	3	3
8	1	2	1	1	3
9	1	2	1	1	3
10	1	2	1	1	3

TIME LINE



FLOW CHART



Why Feature Selection?

- Reduces dimensionality.
- Removes noise, improves classifier stability.
- Identifies key indicators influencing faculty performance.
- Improves interpretability.

CHI-SQUARE FEATURE SELECTION

- **Purpose:** To determine the dependency (relationship strength) between each categorical feature and the target variable (Average of all 32 columns in each row) for feature ranking.
- **Type:** A statistical test used for categorical data.
- **Principle:** It compares the Observed Frequency (actual counts in the data) with the Expected Frequency (what counts would be if the feature and target were independent).
- **Interpretation:** A higher Chi-Square score indicates a greater deviation from expected independence, meaning a stronger relationship with the target.
- **Requirement:** The Chi-Square test only accepts non-negative data points (like counts or frequencies).

DECISION TREE FEATURE SELECTION

- **Embedded Method:** DT performs feature selection during model learning, unlike filter methods like Chi-Square.
- **Selection Mechanism:** Uses Gini Index to pick features that create the most predictive splits.
- **Handles Non-Linearity:** Captures complex interactions between multiple factors that simple statistical methods may miss.
- **Captures Hierarchical Behavior:** Identifies top-level features that have the strongest impact on classification.
- **Model Consistency:** Since the final classifier is also a DT, DT-selected features align perfectly with the model's logic.
- **Interpretable Ranking:** Produces a clear, understandable feature-importance ranking based on the tree structure.

NEED FOR HYBRID SELECTION

The Problem with Single Methods

- High Feature Redundancy: Student evaluation surveys (like the 28 questionnaire items used) contain many highly correlated attributes. Single methods struggle to distinguish redundant from unique information.
- Limited Scope:
 - Filter Methods (e.g., Chi-Square) only measure linear, statistical dependence.
 - Embedded Methods (e.g., DT) only focus on non-linear interactions and model-specific relevance.

The Hybrid Solution

- Faculyze integrates both Chi-Square (for statistical strength) and Decision-Tree Importance (for non-linear strength).
- Goal: To overcome the limitations of single methods by identifying the most influential features based on both statistical relevance and complex non-linear interactions, leading to a more robust and complete feature set.

HYBRID FEATURE SELECTION FRAMEWORK

Strategic Combination

- Faculyze uses a hybrid feature selection approach.
- Chi-Square: Captures linear statistical dependence.
- Decision Tree (DT): Captures non-linear feature interactions.
- Goal: Combine strengths of both for a balanced, robust feature subset.

Ratio-Based Selection

- Total features fixed at 10.
- Eleven ratios tested: 0:10 to 10:0 (DT : Chi).
- Ensures fair comparison across combinations.

Disjoint Rule

- Features selected by Chi-Square are removed from DT's selection pool.
- Prevents redundancy and encourages diverse features.
- Ensures each selected feature adds unique information.

SELECTED FEATURES

	Chi	DT	Features
0	0	10	[Q20, Q16, Q6, Q4, Q23, Q10, Q1, Q7, Q2, Q14]
1	1	9	[Q16, Q20, Q6, Q4, Q23, Q10, Q1, Q7, Q2, Q14]
2	2	8	[Q16, Q23, Q20, Q6, Q4, Q10, Q1, Q7, Q2, Q14]
3	3	7	[Q16, Q23, Q10, Q20, Q6, Q4, Q1, Q7, Q2, Q14]
4	4	6	[Q16, Q23, Q10, Q24, Q20, Q6, Q4, Q1, Q7, Q2]
5	5	5	[Q16, Q23, Q10, Q24, Q18, Q20, Q6, Q4, Q1, Q7]
6	6	4	[Q16, Q23, Q10, Q24, Q18, Q5, Q20, Q6, Q4, Q1]
7	7	3	[Q16, Q23, Q10, Q24, Q18, Q5, Q19, Q20, Q6, Q4]
8	8	2	[Q16, Q23, Q10, Q24, Q18, Q5, Q19, Q13, Q20, Q6]
9	9	1	[Q16, Q23, Q10, Q24, Q18, Q5, Q19, Q13, Q2, Q20]
10	10	0	[Q16, Q23, Q10, Q24, Q18, Q5, Q19, Q13, Q2, Q20]

DISJOINT FEATURE ENFORCEMENT

The Disjoint Rule

- A strict disjointness constraint was enforced during feature subset construction.
- If a feature was selected by the Chi-Square filter, it was immediately excluded from consideration by the Decision-Tree selector, and vice-versa.

Purpose and Benefit

- Ensures Independent Contribution: This rule eliminates feature overlap, guaranteeing that each of the final 10 features contributes unique and independent information to the prediction.
- Reduces Redundancy: It is critical for highly correlated survey data where many questions might measure the same underlying construct. By forcing disjointness, the framework ensures the final subset is as non-redundant as possible.

MODEL TRAINING

Data Setup

- Dataset: Türkiye Student Evaluation Dataset.
- Splitting: The dataset was partitioned using stratified sampling.
 - Train/Test Split: 80% used for model training; 20% reserved for unbiased testing.
 - Stratification: Ensures all three performance classes are represented proportionally in both sets, mitigating class imbalance effects.

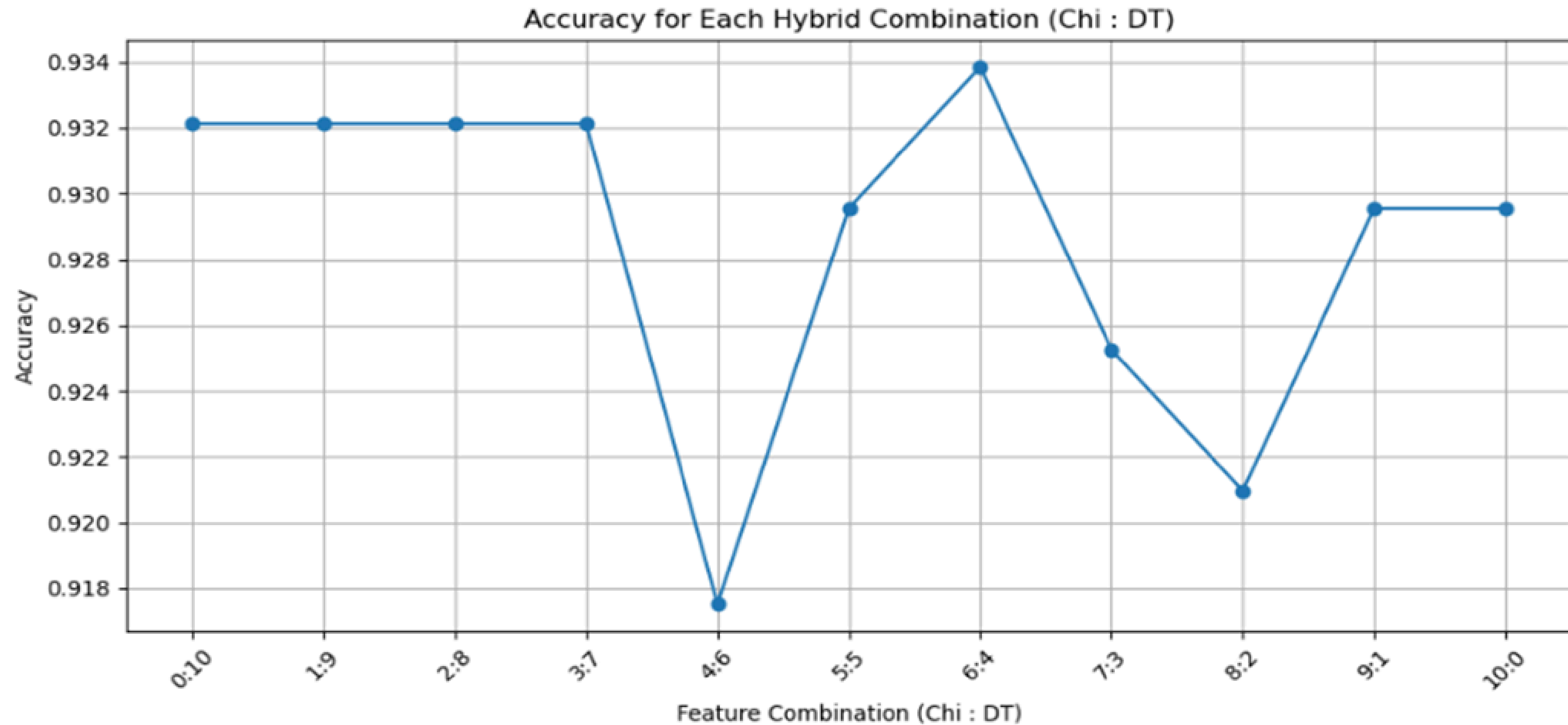
The Experiment

- 11 Different Models were run—one for each of the eleven unique hybrid feature subsets.
- Each subset was used to train the Decision Tree, and its performance was recorded.
- The Best-Performing Model was chosen based on achieving the highest combined scores across all evaluation metrics.

EVALUATION METRICS

- **Accuracy:** How correctly the model predicts overall.
- **Precision:** How careful the model is with positive predictions.
- **Recall:** How well the model captures actual true cases.
- **F1-Score:** Balanced measure of precision and recall.
- **Specificity:** How well the model identifies true negatives

ANALYSIS OF ALL COMBINATIONS



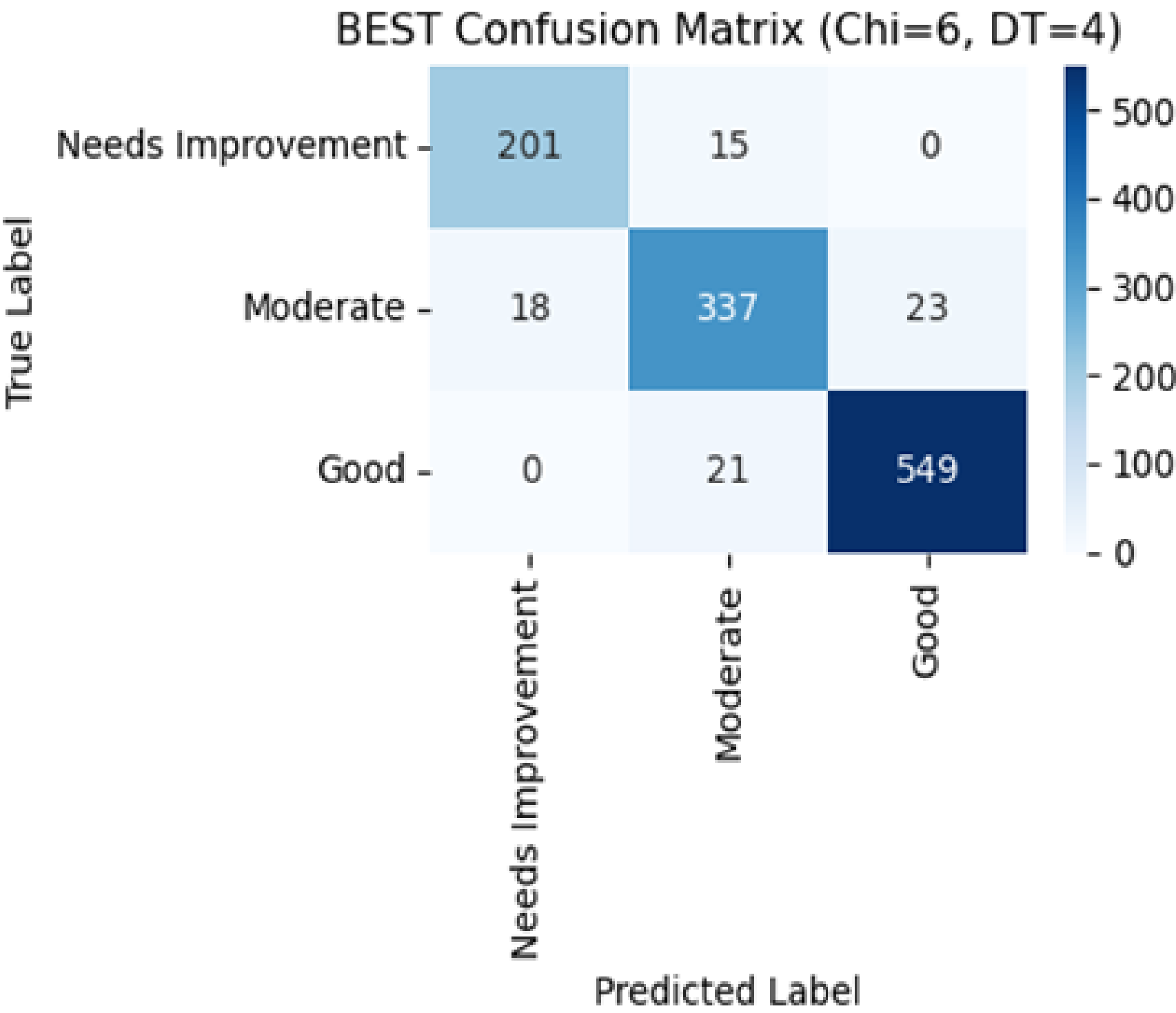
RESULTS OVERVIEW

- Accuracy changed for nearly each Chi-DT ratio.
- Hybrid ratios show different strengths.
- Some ratios overfit while others generalize better.
- DT-heavy combinations generally performed better.
- Hybrid approach is more stable than individual methods.

BEST HYBRID RATIO IDENTIFIED

Hybrid Ratio (Chi: DT)	Number of Features	Accuracy	Precision	Recall	F1	Specificity
0: 10	10	93.21%	93.22%	93.21%	93.22%	96.48%
1: 9	10	93.21%	93.22%	93.21%	93.22%	96.48%
2: 8	10	93.21%	93.20%	93.21%	93.20%	96.43%
3: 7	10	93.21%	93.20%	93.21%	93.20%	96.43%
4: 6	10	91.75%	91.72%	91.75%	91.73%	95.57%
5: 5	10	92.96%	92.94%	92.96%	92.95%	96.31%
6: 4	10	93.38%	93.37%	93.38%	93.38%	96.55%
7: 3	10	92.53%	92.48%	92.53%	92.49%	95.97%
8: 2	10	92.10%	92.04%	92.10%	92.06%	95.80%
9: 1	10	92.96%	92.96%	92.96%	92.96%	96.29%
10: 0	10	92.96%	92.96%	92.96%	92.96%	96.29%

BEST MODEL CONFUSION MATRIX



CONCLUSION

- The hybrid feature-selection approach combining Chi-Square and Decision Tree methods improved the accuracy and reliability of faculty-performance prediction.
- Enforcing disjoint feature sets helped us understand the unique contribution of each method and avoided feature overlap, making the evaluation fair and interpretable.
- The systematic testing of all ratios (0:10 to 10:0) showed that Decision-Tree-based features capture most of the meaningful patterns in student feedback.
- Overall, Faculyze provides a structured, data-driven framework that institutions can use to identify important teaching-quality indicators and make better evaluation decisions.

FUTURE SCOPE

- Explore advanced machine-learning models such as ANN, Random Forest, SVM, Logistic Regression to compare performance.
- Try deep-learning-based architectures like LSTM, Bi-LSTM, or lightweight neural networks for more complex feedback patterns.
- Use additional feature-selection techniques like ReliefF, Mutual Information, Recursive Feature Elimination (RFE), or Genetic Algorithm-based selection.
- Test the system on real university datasets to improve generalization and validate results in practical environments.
- Deploy Faculyze as a fully functional web application for institutions to evaluate faculty performance in real time

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THANK YOU